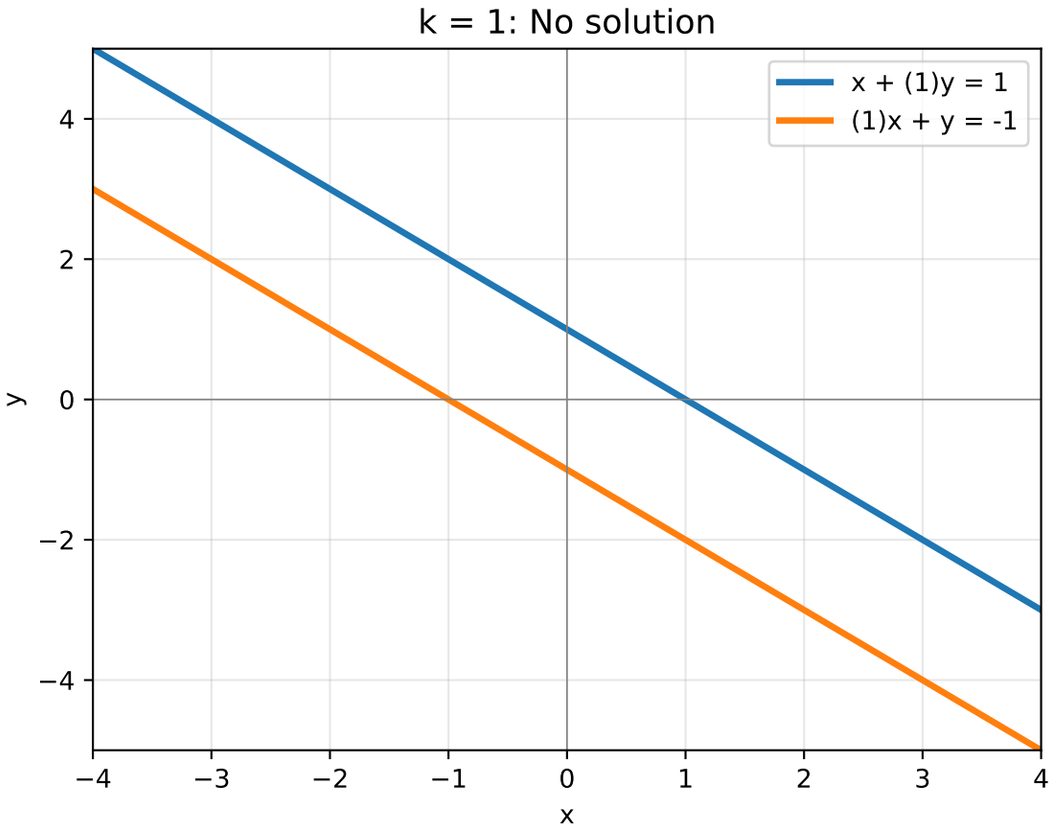
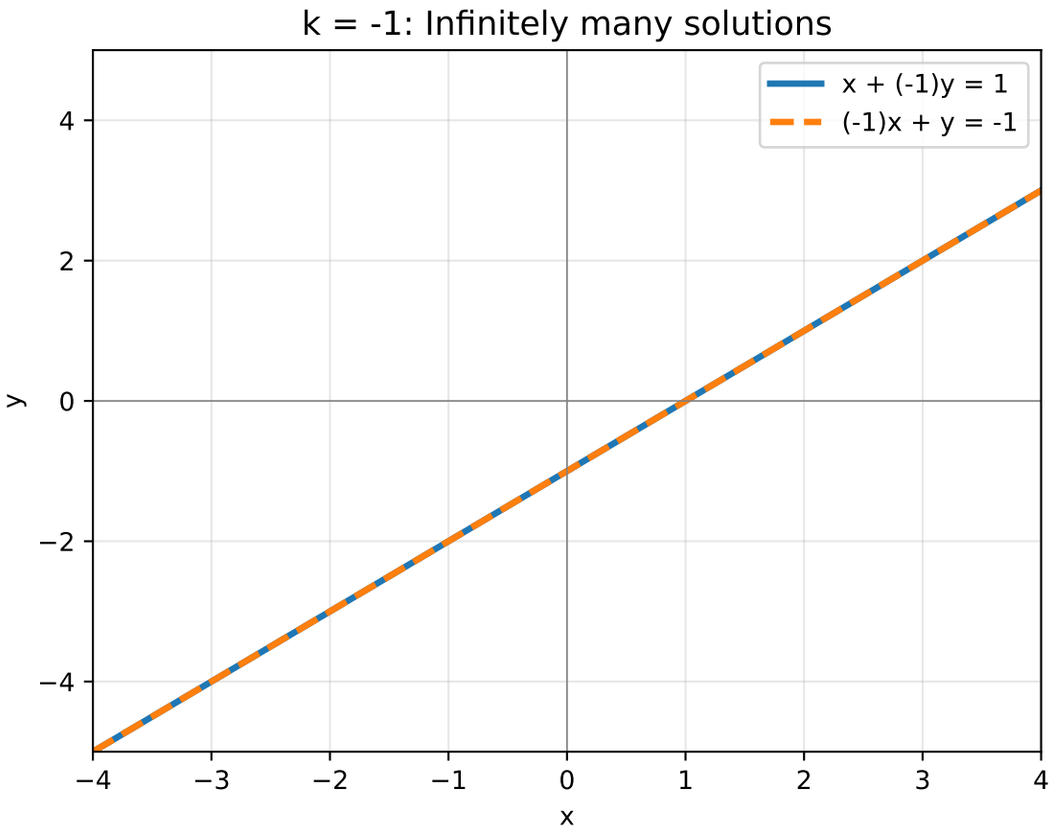
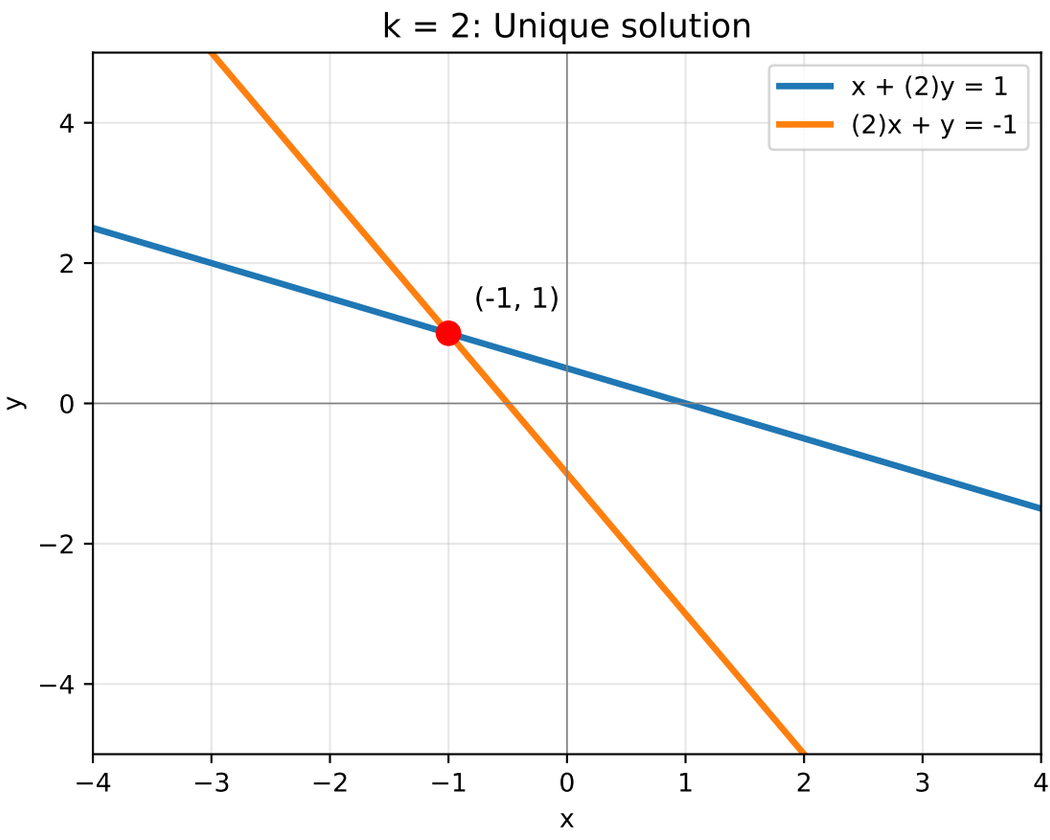

k = 1: No solution
x + (1)y = 1
(1)x + y = -1
x
y


k = -1: Infinitely many solutions
x + (-1)y = 1
(-1)x + y = -1
x
y


k = 2: Unique solution
x + (2)y = 1
(2)x + y = -1
(-1, 1)
x
y